

CSA0614

DESIGN AND ANALYSIS OF ALGORITHMS

Name: Haresh Kumar N L

Reg No: 192425009

Experiment 14: Exponentiation

Code:

```
def power(x, n):  
    if n == 0:  
        return 1  
    half = power(x, n // 2)  
    if n % 2 == 0:  
        return half * half  
    else:  
        return x * half * half  
  
x = 2  
n = 10  
print("Result:", power(x, n))
```

Output:

```
===== RESTART: C:/Us  
Result: 1024
```